Semester	T.E. Semester VI – CMPN
Subject	DevOps Lab
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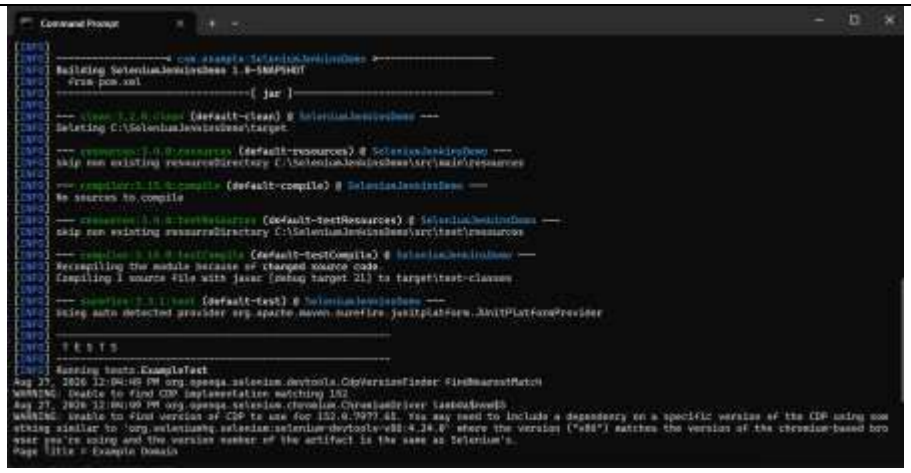
Grade and Subject Teacher's Signature		

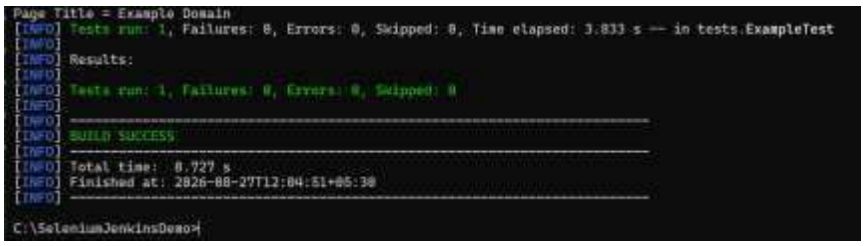
Experiment Number	06
Experiment Title	Setup and Run Selenium Tests in Jenkins Using Maven
Objectives (Skill Set / Knowledge Tested / Imparted)	The primary **objective** of this experiment is to automate software quality assurance by integrating **Selenium automated UI tests** into a **Jenkins Continuous Integration (CI)** pipeline using **Maven**. It aims to teach developers and QA engineers how to transition tests from a local manual execution environment to an automated, headless browser environment triggers on demand.
Description	This lab experiment guides users through a comprehensive, step-by-step setup divided into seven structured phases. It begins with installing core prerequisites like **Java JDK 21**, **Maven**, and **Google Chrome**, followed by building a Maven project integrated with **JUnit 5** and **Selenium WebDriver**. The tests are intentionally configured to run in **headless mode** to ensure compatibility with server environments that lack a graphical user interface. Finally, the project is wired into a **Jenkins Freestyle job**, where execution is automated and **JUnit XML test reports** are generated and published directly to the Jenkins dashboard for clear tracking of build success.
Program	**Creating the Maven Selenium Project and running Selenium using Maven first :**

```
Command Prompt

    ├──surefire-reports
    │       TEST-tests.ExampleTest.xml
    │       tests.ExampleTest.txt
    │
    └──test-classes
        └──tests
                ExampleTest.class

C:\SeleniumJenkinsDemo>notepad pom.xml

C:\SeleniumJenkinsDemo>notepad src\test\java\tests\ExampleTest.java

C:\SeleniumJenkinsDemo>notepad src\test\java\tests\ExampleTest.java

C:\SeleniumJenkinsDemo>cd C:\SeleniumJenkinsDemo

C:\SeleniumJenkinsDemo>dir
 Volume in drive C is OS
 Volume Serial Number is D8C5-D4C2

 Directory of C:\SeleniumJenkinsDemo

27-08-2026  12:03    <DIR>          .
27-08-2026  11:54             1,484 pom.xml
27-08-2026  11:45    <DIR>          src
27-08-2026  11:55    <DIR>          target
               1 File(s)          1,484 bytes
               3 Dir(s)  339,548,237,824 bytes free

C:\SeleniumJenkinsDemo>mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:SeleniumJenkinsDemo >-------------------
[INFO] Building SeleniumJenkinsDemo 1.0-SNAPSHOT
```

Starting Jenkins:

```
Command Prompt

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kirti>cd C:\

C:\>mkdir SeleniumJenkinsDemo\src\test\java\tests
A subdirectory or file SeleniumJenkinsDemo\src\test\java\tests already exists.

C:\>cd SeleniumJenkinsDemo

C:\SeleniumJenkinsDemo>tree /F
Folder PATH listing for volume OS
Volume serial number is 00000062 D8C5:D4C2
C:.
│   pom.xml
│
├───src
│   └───test
│       └───java
│           └───tests
│                   ExampleTest.java
│
└───target
    ├───generated-test-sources
    │   └───test-annotations
    ├───maven-status
    │   └───maven-compiler-plugin
    │       └───testCompile
    │           └───default-testCompile
    │                   createdFiles.lst
    │                   inputFiles.lst
    │
    ├───surefire-reports
    │       TEST-tests.ExampleTest.xml
    │       tests.ExampleTest.txt
    │
```



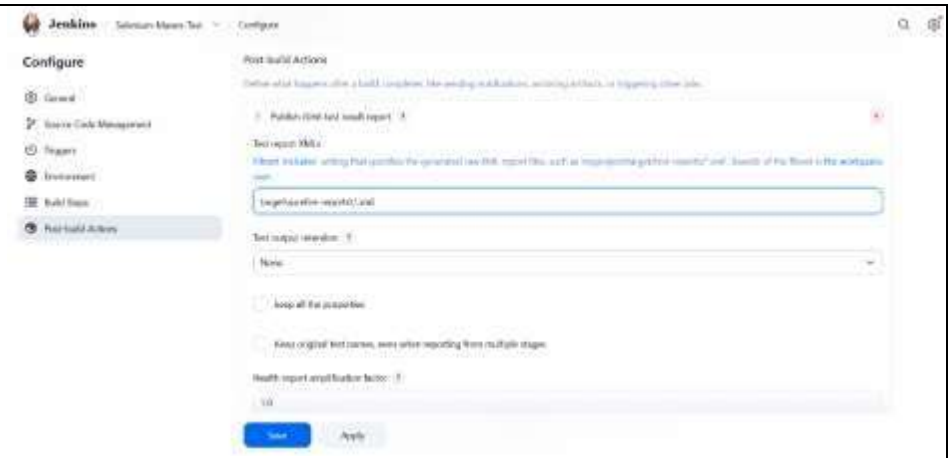


Configure Java and Maven inside Jenkins:

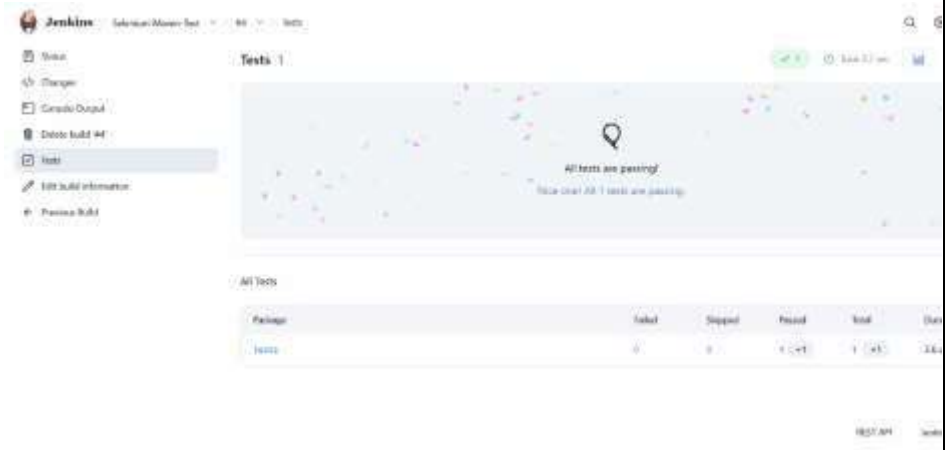
Build Steps
Invoke top-level Maven targets
Maven Version
Maven3
Goals
clean test
Advanced
Add build step

Jenkins
Maven installations
Add Maven
Maven
Name
Maven3
MAVEN_HOME
Install automatically
Add Maven
Save
Apply

Jenkins



Selenium Test Results in Jenkins:



Conclusion	The integration of Selenium, Maven, and Jenkins establishes a robust Continuous Integration workflow that enables seamless headless test execution in server environments. Publishing JUnit XML test reports to the Jenkins dashboard ultimately provides development teams with immediate feedback on application stability and build health. You can read the full guide on the project website.